
Optimizing the Extrusion Machine Parameters for PLA Using Full Factorial Design Approach



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ABSTRACT

It is the case of this report which seeks to analyse and determine the various extrusion machine parameters for the production of high quality PLA filaments through the implementation of full factorial design of experiments DOE. PLA, a bio-degradable polymer coming from renewable resources is well adapted in 3D prints and in other uses. The work highlights the consequences of die temperature, roller speed, spindle speed, and inlet temperature on filament size and has used Minitab to analyze the results statistically. The regression model and ANOVA results further suggest that the key parameters affecting the size of the filament include spindle speed and roller speed while die temperature has the moderate, and inlet temperature had the least effect. This work contributes to optimising the parameters for PLA filaments synthesis, reducing variability and improving the properties of the final material, thus promoting sustainable production. Also, it was discovered that the full factorial DOE approach has proved useful in determining both the main and interaction effects while giving a sound foundation for future research on extrusion technology. This work also optimized the production of PLA filament while at the same time addressing green manufacturing goals.

INTRODUCTION

1.1. PLA Extrusion

Plastic extrusion is one of the main technologies used to produce plastic products. As a result of this process, so-called continuous products in the form of profiles, plates, or pipes are obtained. This technology is also used to manufacture composite products such as reinforced hoses for transporting liquids or gases and electric wires. This process also produces packaging [[1,2,3,4,5,6](#)].

This process is quite well known and described, but research is still being carried out on modeling this process, both for the plastics themselves and those filled with organic fillers (wood flour, cellulose fibers) and mineral fillers (calcium carbonate, glass fibers) [[6,7,8,9,10](#)], among other substances. To sum up, this process consists of plasticizing the material in the plasticizing system of the extruder, and then shaping it by passing it through a properly shaped head, which is an open tool [[11,12](#)].

Plastics in the form of granules, powders, flakes, or chips are used as the starting material. Due to the technological process itself, the plasticizing mechanism and its rheological properties, various screw designs as well as boring heads are used, in which appropriate changes in geometry are introduced [[13,14,15,16,17,18,19,20](#)].

Poly lactide (PLA) was used in the study of the extrusion process. It is classified as biodegradable plastic and is a promising polymer for this type of application. It can be produced annually with a capacity of more than 140,000 tons. Low greenhouse gas emissions and low energy consumption are also notable factors [[21,22,23](#)]. Substances of mineral origin, such as calcium carbonate and others, can be successfully added to PLA as fillers [[24,25,26,27,28](#)].

Products made from this material can be produced using the technology of injection, extrusion, and 3D printing [[22,23,24,25](#)]. Compared to typical polymers intended for these production technologies, it is characterized by a higher density, ranging from 1.20 g/cm³ to about 1.35 g/cm³ depending on the type. Typical plastics (PP, PE, PS, ABS, and others) have a density of 0.9 g/cm³ to 1.05 g/cm³. This parameter is not without influence on the processing and production of products, both in the technology of injection, extrusion, or at least the increasingly popular 3D printing. In this case, this material is one of the most popular raw materials [[29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44](#)].

This paper attempts to assess the PLA extrusion process depending on the process conditions, such as screw rotational speed and temperature. Studying the impact of these parameters on the extrusion process is important, both from the point of view of optimization of the production process, and the future development of computer models for designing and simulating this process. The study of the impact of screw rotational speed and temperature on the PLA extrusion process aims to understand how these variables affect the quality and properties of the final product. For example, temperature has a key influence on a plastic's viscosity, flowability, and other rheological properties that affect the process.

Literature analysis in this area provides important information on the optimal ranges of temperatures and rotational speeds for traditional polymers (PE, PP), which lead to the best results in the extrusion process. The results obtained in this work can be used to develop a computer model that will predict the behavior of PLA under various conditions of this technological process. Such a model can be extremely valuable when designing new production processes, optimizing existing ones and anticipating potential problems.

The undoubted novelty of this development is the comprehensive approach that takes into account both the extruder system and the head tool.

1.2. Objectives

There are the following objectives that are to be aimed for this report:

- To investigate the influence of process parameters on the quality of PLA filament.
- To identify significant factors and interactions affecting extrusion outcomes.
- To determine the optimal combination of parameters for achieving consistent filament diameter and mechanical properties.

1.3. Extrusion Machine

An extrusion machine is a manufacturing tool that forms objects having a predetermined cross-sectional shape by passing the material through a die. Extrusion means forcing raw material including plastic, metals, rubber or foods through a die of specific cross-sectional shape resulting in profile of uniform section through its length and cutting it to the required length. Extrusion machine components include the hopper which feeds the materials, feed screw which conveys the

material forward, barrel, which melts the material by rotating the screw, and the die which forms the cross-section of the extruded product. Based on the flow of material through the die, after coming out of the die chamber the material is cooled and hardened by air or water, a puller ensuring uniform shape and length of the product while a cutter customizes it to the desired length.

The extrusion machines can be used for both hot and cold extrusion and the decision taken depends on whether the material used for extrusion is heated or not. For metals and in cases of plastics, hot extrusion is employed while cold extrusion is appropriate for metals and high strength parts. The age-old plastic extrusion and processing industries use these machines to manufacture pipes, sheets, films, and profiles while the food processing and manufacturing companies use them to prepare snacks; breakfast cereals and similarly, other processed foods and related products. The extrusion machines are flexible in operation, accurate and are capable of creating various cross-sectional structures and profiles without interruption, thus play an important role in manufacturing industries, aerospace, automobile industries and food industries.

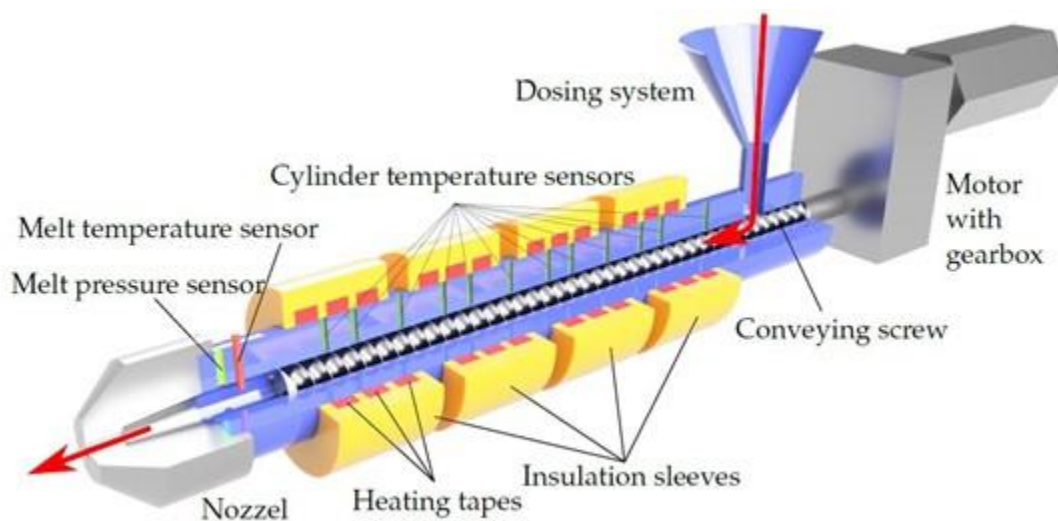


Figure 1 Screw Type Extrusion Machine

1.3.1. Types of Extrusion Machines

There are following types of extrusion machines:

1. Single Screw Extruder.
2. Twin Screw Extruder
3. Hot Extrusion Machine

-
4. Cold Extrusion Machine
 5. Hydraulic Extruder
 6. Vertical Extruder
 7. Film and Sheet Extruder

1.3.1.1. Single Screw Extruder

This is the most standard type of extrusion machine for the processing of plastic material and the like. It employs a single helical screw that conveys the material, melts the material as it is transported through a heated barrel then forces it through a die. The single screw extruder is extensively used for the production of plastic films, sheets, and pipes etc.

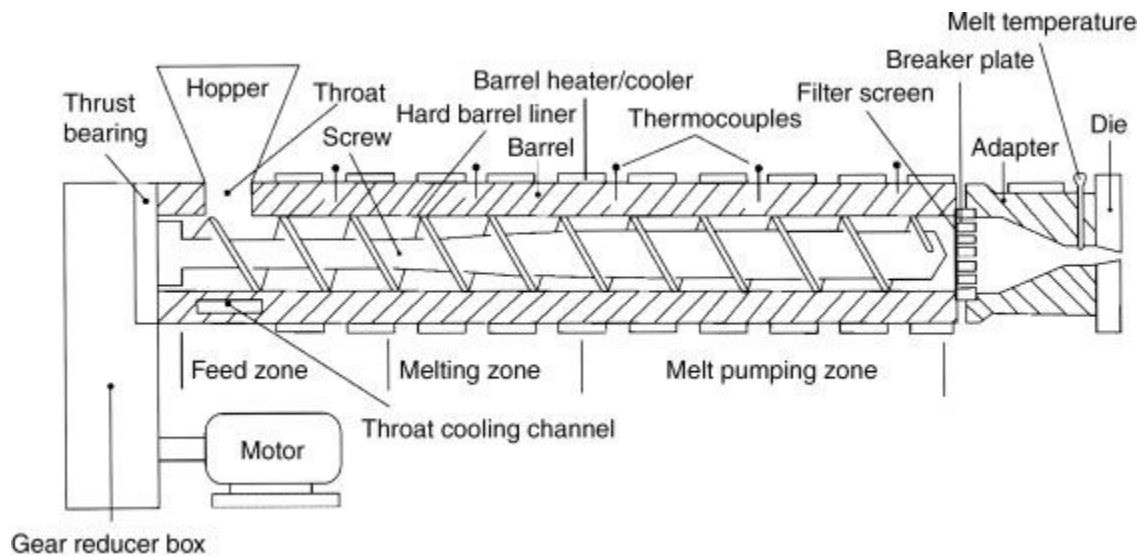


Figure 2 Single Screw Extrusion machine

1.3.1.2. Twin Screw Extruder

In a twin screw extruder the barrel contains two interlocking screws that rotate in a direction against the other. However, this design enables a better mixing and shearing of the material, it may be ideal for complex or sensitive material. The twin screw extruders are widely employed in the compounding process, polymer blending, and manufacturing of functionally graded polymers and

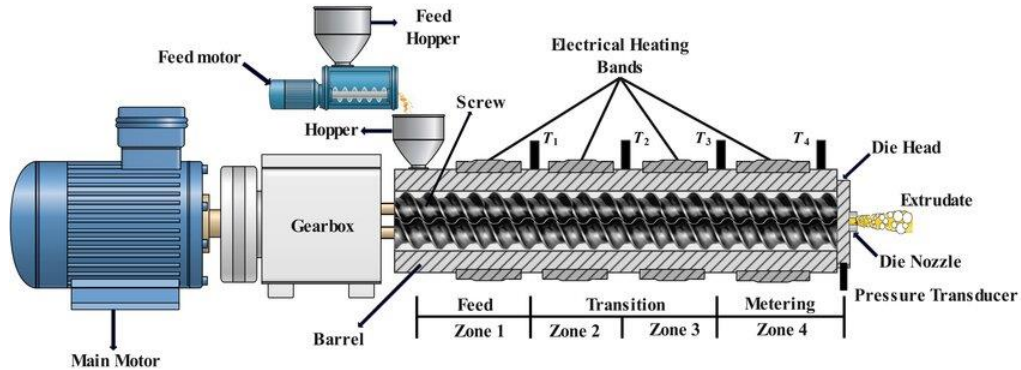


Figure 3 Twin screw extrusion machine

foods.

1.3.1.3. Hot Extrusion Machine

In hot extrusion the material is first heated before it is forced through the die opening. This is used for metals, plastics and rubber etc in practice to remove the gas evolved during the reaction. This is made by forcing the heated material through a die to produce continuous cross sections like a round shape (rods, bars or profiles). Hot extrusion is regularly applied in metal working process in which the aluminum, copper as well as steel goes through extrusion procedures.

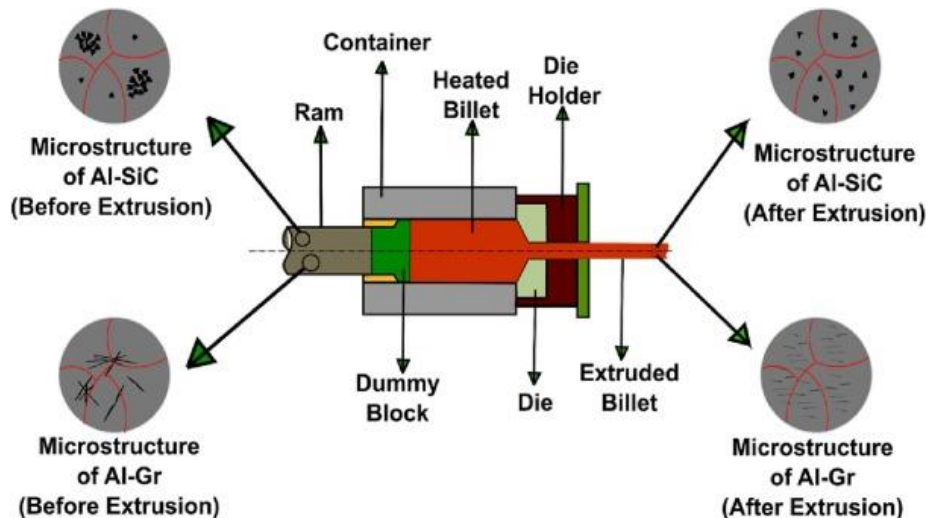


Figure 4 Hot Extrusion machine process

1.3.1.4. Cold Extrusion Machine

Cold working method involves use of pressure without involving heat to change the cross-sectional area of the material. This method is mainly applied for metals especially for high strength or when

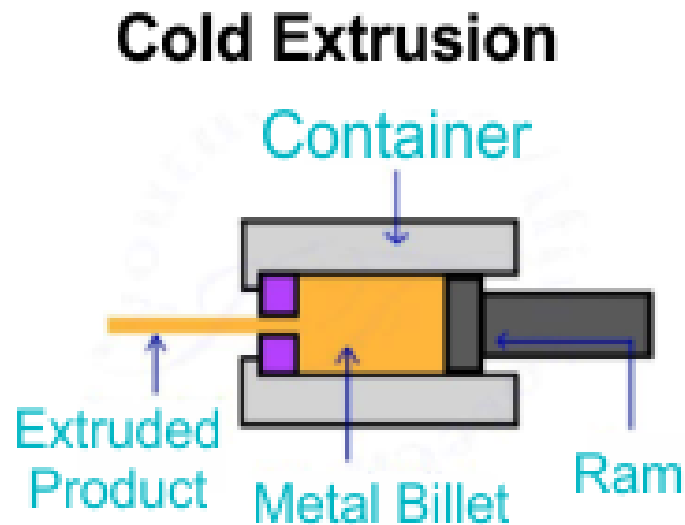


Figure 5 Cold Extrusion process

the metal requires to retain some mechanical properties. Cold extrusion is widely used in the field of automotive, Aerospace and manufacturing industries for the manufacturing of superior accuracy parts.

1.3.1.5. Hydraulic Extruder

Hydraulic extrusion uses hydraulic pressure to force the material through the die opening of the hydraulic extrusion machines. This kind of machine is most suitable in industries that apply high levels of force during extrusion and most suitable for metals such as steel and aluminum.

1.3.1.6. Vertical Extruder

A vertical extruder, in particular, has a vertical barrel and die. It is aimed at products which have to be extruded into long vertical shapes, for example, large tubes made of plastics or some metal articles. The high speed options of this machine type are more frequently applied in the plastic and rubber industry.

1.3.1.7. Film and Sheet Extruder

This kind of an extrusion machine is specifically used in manufacturing of films and sheets of materials such as plastics. The extruder takes the material through a die and after that uses rollers to develop the sheet or the film into several thin and even sub layers. Film and sheet extruders serves packaging, textile and automotive industries.

Table 1 Comparison between types of extrusion machines

Type of Extrusion Machine	Design	Common Materials	Applications	Advantages	Typical Industries
Single Screw Extruder	Single rotating screw in a barrel	Plastics (PVC, PE, PP), rubber, food	Plastic films, sheets, pipes, profiles	Simple design, cost-effective, high output	Packaging, automotive, plastic manufacturing
Twin Screw Extruder	Two intermeshing screws in a barrel	Plastics, food, composites, polymers	Polymer blending, compounding, food processing	Better mixing, precision, high shear forces	Polymer compounding, food, pharmaceuticals
Hot Extrusion Machine	Heated barrel with a single or twin screw	Metals (aluminum, steel, copper), plastics	Metal rods, bars, profiles, plastic products	Suitable for high-strength materials, versatility	Metalworking, automotive, aerospace
Cold Extrusion Machine	No heat, uses high pressure	Metals (steel, aluminum, copper)	High-precision metal components, fasteners	No heating required, maintains material properties	Aerospace, automotive, manufacturing

Hydraulic Extruder	Hydraulic pressure to push material	Metals, some plastics	Large force extrusion of metals, plastic profiles	High pressure capability, suitable for tough materials	Metalworking, aerospace, heavy industries
Food Extrusion Machine	Varies (screw extruder or others)	Food ingredients (corn, rice, wheat, starch)	Snacks, breakfast cereals, pasta, pet food	High efficiency, easy to scale production	Food processing, snack production, pet food
Vertical Extruder	Vertical barrel and die	Plastics, rubber, some metals	Large vertical profiles (pipes, rods)	Suitable for large-scale vertical products	Plastics, rubber, pipe manufacturing
Film and Sheet Extruder	Flat die with rollers	Plastics (PE, PP, PET), some rubber	Thin plastic sheets and films	Produces thin, uniform sheets/films	Packaging, automotive, textiles

1.4. Full Factorial Design of Experiment

Factorial Design of Experiment (DOE) is a technique of working interacting factor on two or more than two variables at a time. They are each scored at a number of levels and all variants of these levels are compared and analysed. It is beneficial in the sense that it enables a researcher to capture both main effects of individual factors, and interactions effects. For instance, in a case of product development a factorial design can accustom itself to the behaviour of price, features and marketing techniques.

One advantage of using factorial designs is that they are relatively quick and efficient to implement to measure multiple variables in a single survey rather than singular ones. This leads to better understanding of the relationship impact of one factor with others which is important if the relationship between these factors have a strong impact on the result. Nevertheless, the complexity

and the resource demands associated with it rise in direct proportion to the number of factors and levels, which may complicate the design for experiments, where many factors are applied.

Uses of factorial design are found in different disciplines, for example, production, selling, medical and health sectors, and research among others, where they are used in solving systematic issues. Because of its flexibility and comprehensive procedure, it provides solid analysis, which is why experimental analysis methods use it as a foundation.

METHODOLOGY

2.1. Full Factorial Design of Experiment

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2.2. Software Used

For performing the experiments Minitab software is used because it can provide with variety of options with accurate results. Minitab is one of the most popular and efficient statistical programs for the analysis of factorial designs because it includes a wide range of tools for Design of Experiments (DOE). It can generate full and fractional factorial designs more easily since researchers have platforms where they can define factors, levels, and replicates easily. This eliminates some of the challenges that comes along with the manual setting up of experiments so that there can be better control of the design generation.

Of all the features that Minitab has to offer, it can handle interaction, which is very vital in factorial designs. For instance, it can find out whether the impact of one factor, say temperature, is

influenced by the second factor, that is, the extrusion speed. This evaluation is done with statistical reliability employing analysis tools such as Analysis of Variance (ANOVA), which can help the end user to determine which factors/interactions have an impact on the result.

The understandability of results can be facilitated with Minitab's interaction plots, main effects plots and response surface plots. It is from such plots the various factors that affect the response variable can easily be explained, and thus helping greatly in decision making to enhance the process.



Figure 6 Minitab software logo

2.3. Experimentations

There are the following design considerations for it:

- No. of factors affecting response variables.
- No. of Response variables.
- No. of levels this design will be having.

This system is a 4 factored 2 leveled design with a single response variable named as “Filament Size”.

Minitab is used in this experiment to discover the main effects for the parameters that involve in the filament fabrication process. Analysis of Variance (ANOVA) remained proficient in finding the conditional on variables that affect the machining appearances and to create the optimum situation. There is four parameter use in this experiment, Table 2 shows the range of parameter factor and level were carried out in two level high and low. The high was design in (+1) and for low was designed in (-1). The response is very important because from that the result of this experiment will be obtained. The incline of the factor that continued in demand to recognized the

response which is the filament wire size. Two level full factorial design have four factors (A, B, C and D) and each factor to be studied at 2-level. There are altogether 24 runs with added 8 center points per blocks.

Table 2 Factors and their levels to be designed

Factor code	Name	Units	Low (-1)	High (+1)
A	Die temperature	°C	100	160
B	Roller puller speed	rpm	2	6
C	Spindle speed	Hz	2	6
D	Inlet temperature	°C	60	100

Table 3 Full factorial experiment runs

Sr. No.	Die Temperature	Roller Speed	Spindle Speed	Inlet Temperature
1	100	6	6	80
2	100	2	2	80
3	100	6	2	80
4	160	6	6	50
5	160	6	6	80
6	100	6	2	50
7	100	2	6	50
8	100	2	2	50
9	160	2	2	50
10	160	2	6	80
11	160	6	2	80
12	160	2	2	80
13	100	2	6	80
14	160	2	6	50
15	160	6	2	50
16	100	6	6	50

RESULTS

The complete result of all 16 runs of experiment are shown in Table 3. From the experiment, it can be seen that there are two result that produce the filament with diameter size in range of 1.75mm-1.80 mm which are run number 4 and 5. The parameter setting is when the die temperature at 160°C, the roller puller speed at 6 RPM, the spindle speed at 6 Hz. The differences is only at the inlet temperature where at run 4 the temperature is at 50°C meanwhile at run 5 the temperature is at 80°C.

Table 4 The respond result for filament size diameter

Sr. No.	Die Temperature (Factor A)	Roller Speed (Factor B)	Spindle Speed (Factor C)	Inlet Temperature (Factor D)	Filament Size (Response)
1	100	6	6	80	1.63
2	100	2	2	80	1.18
3	100	6	2	80	0.93
4	160	6	6	50	1.77
5	160	6	6	80	1.76
6	100	6	2	50	0.98
7	100	2	6	50	2.49
8	100	2	2	50	1.43
9	160	2	2	50	1.70
10	160	2	6	80	2.73
11	160	6	2	80	1.11
12	160	2	2	80	1.36
13	100	2	6	80	1.98
14	160	2	6	50	2.81
15	160	6	2	50	1.12
16	100	6	6	50	1.53

2.3.1. General Factorial Regression

2.3.1.1. Factor Information

Table 5 Factors information for experimentations

Factor	Levels	Values
Die Temperature	2	100, 160
Roller Speed	2	2, 6
Spindle Speed	2	2, 6
Inlet Temperature	2	50, 80

2.3.1.2. Regression Equation

$$\text{Filament Size} = 1.115 + 0.00460 \text{ Die Temperature} - 0.1516 \text{ Roller Speed} + 0.2153 \text{ Spindle Speed} - 0.00479 \text{ Inlet Temperature}$$

2.3.1.3. Coefficients

Table 6 Coefficients for experimentations

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	1.115	0.340	3.28	0.007	
Die Temperature	0.00460	0.00166	2.77	0.018	1.00
Roller Speed	-0.1516	0.0249	-6.08	0.000	1.00
Spindle Speed	0.2153	0.0249	8.64	0.000	1.00
Inlet Temperature	-0.00479	0.00332	-1.44	0.177	1.00

2.3.1.4. Model Summary

Table 7 Model Summary for experiments

S	R-sq	R-sq(adj)	R-sq(pred)
0.199469	91.68%	88.66%	82.41%

2.3.1.5. Analysis of Variance

Table 8 ANOVA Analysis for experiments

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	4.82508	1.20627	30.32	0.000
Die Temperature	1	0.30526	0.30526	7.67	0.018
Roller Speed	1	1.47016	1.47016	36.95	0.000
Spindle Speed	1	2.96701	2.96701	74.57	0.000
Inlet Temperature	1	0.08266	0.08266	2.08	0.177
Error	11	0.43767	0.03979		
Total	15	5.26274			

2.3.1.6. Graphical representation

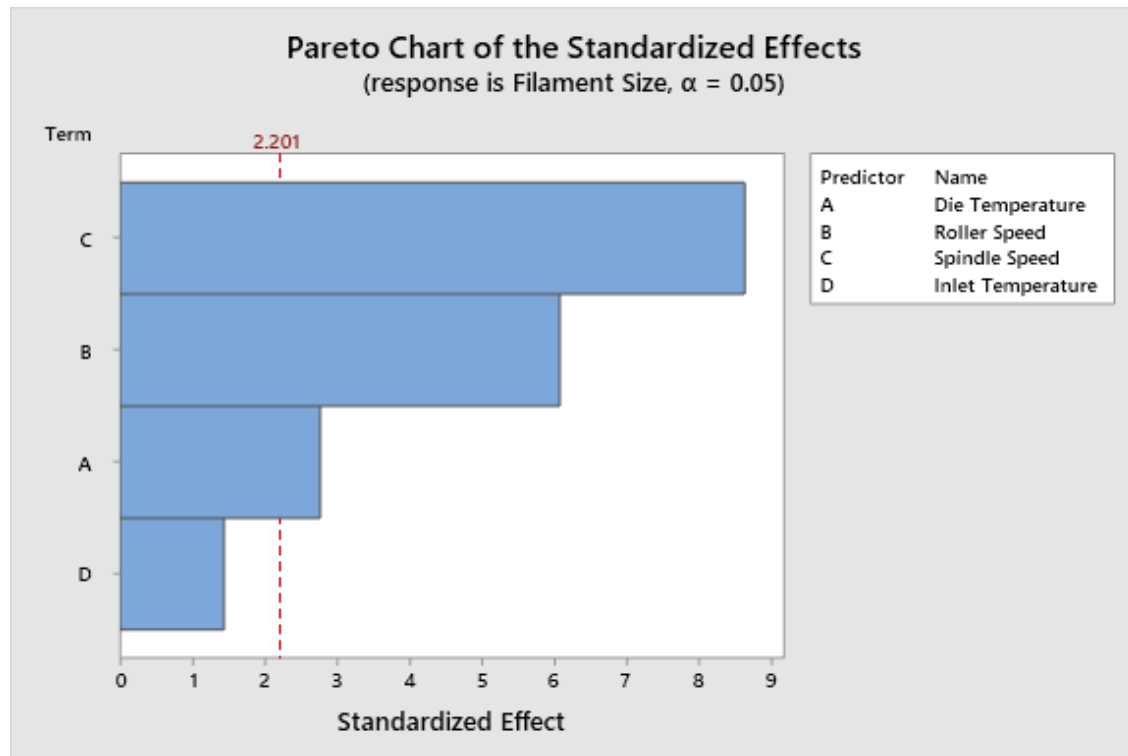


Figure 7 Pareto Analysis for experimentations

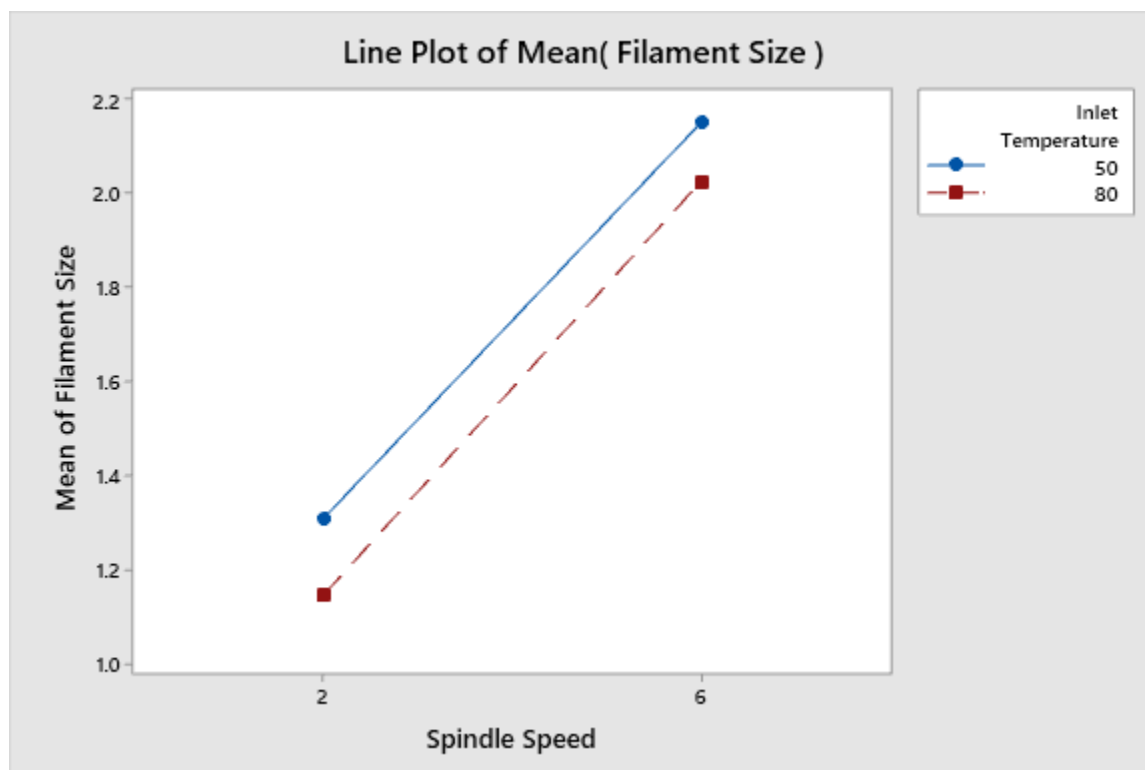


Figure 8 Line plot for mean filament size VS Inlet Temperature VS Spindle Speed

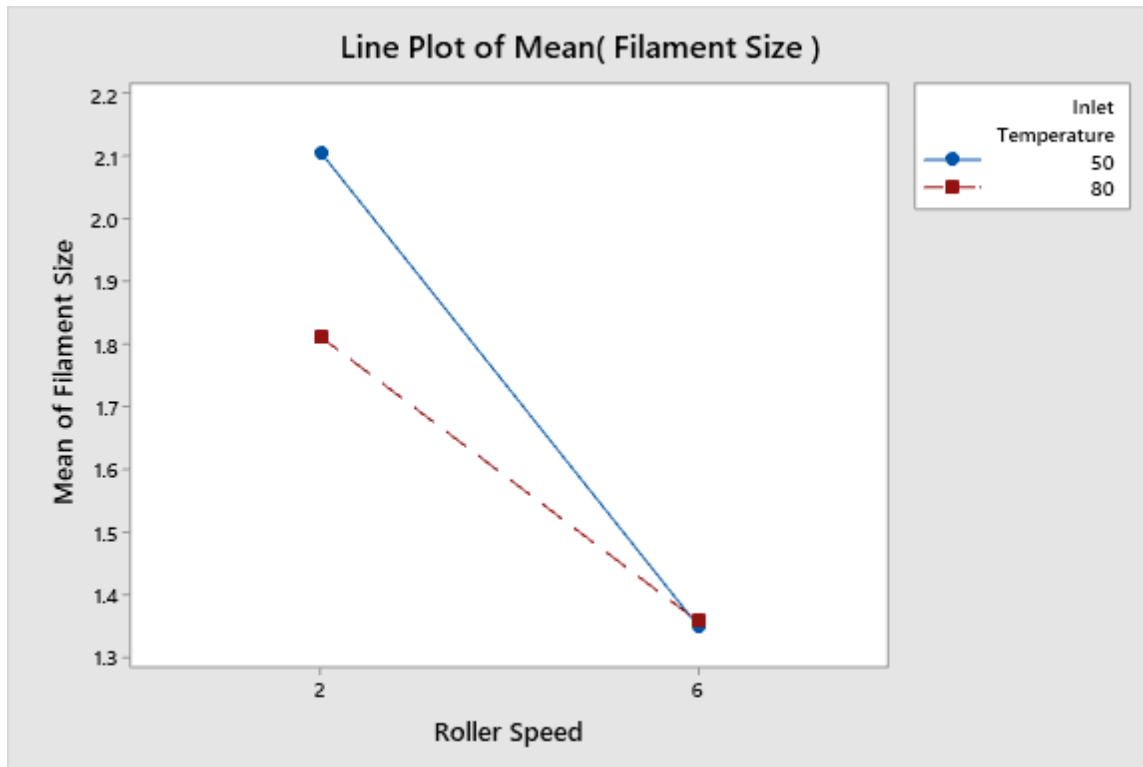


Figure 9 Line plot for mean filament size VS Inlet Temperature VS Roller Speed

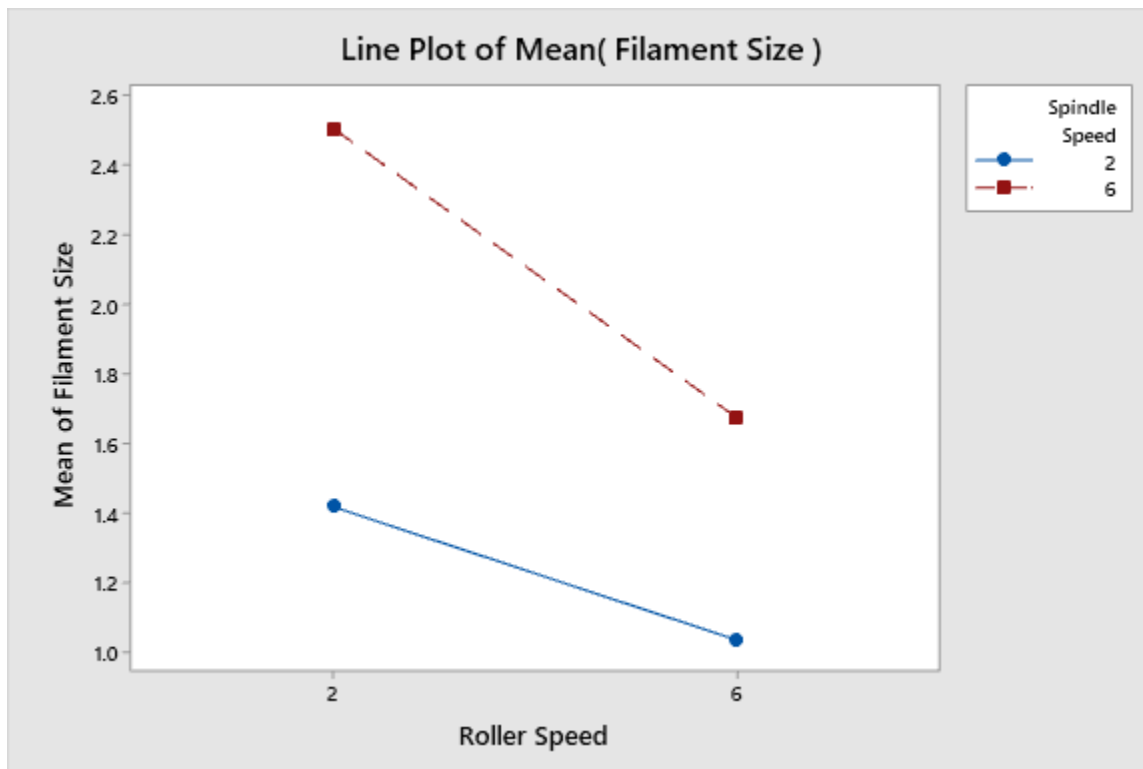


Figure 10 Line plot for mean filament size VS Roller Speed VS Spindle Speed

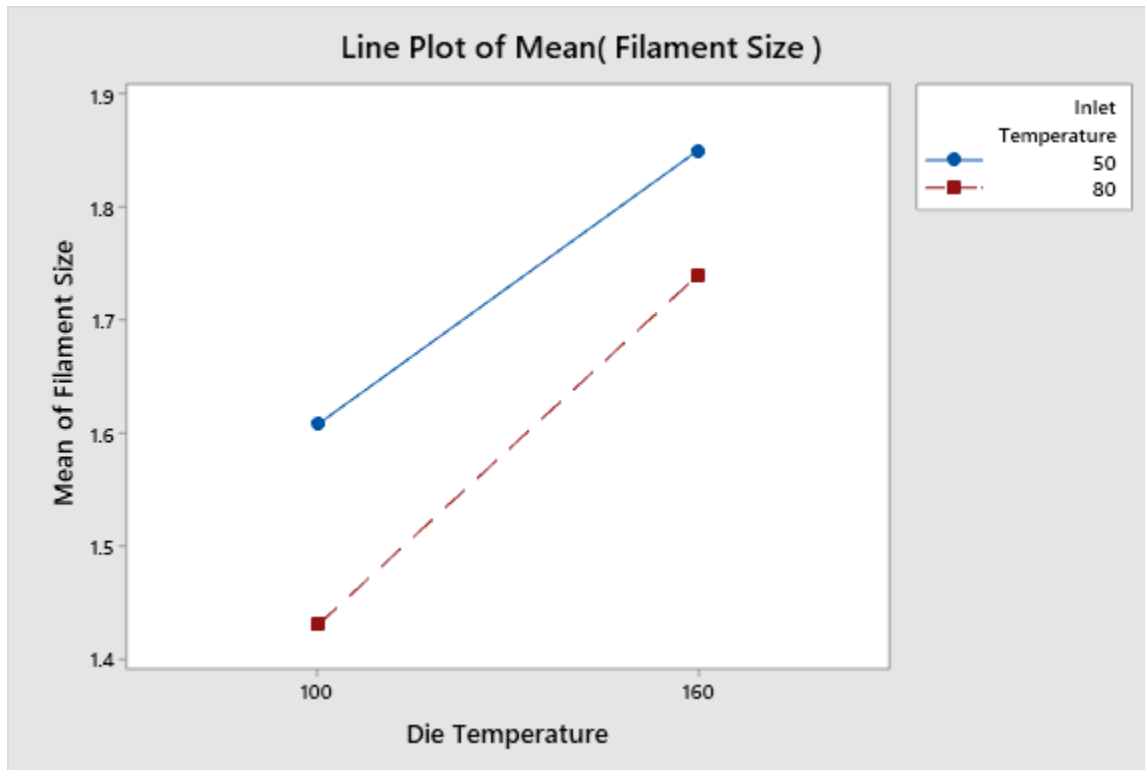


Figure 11 Line plot for mean filament size VS Inlet Temperature VS Die Temperature

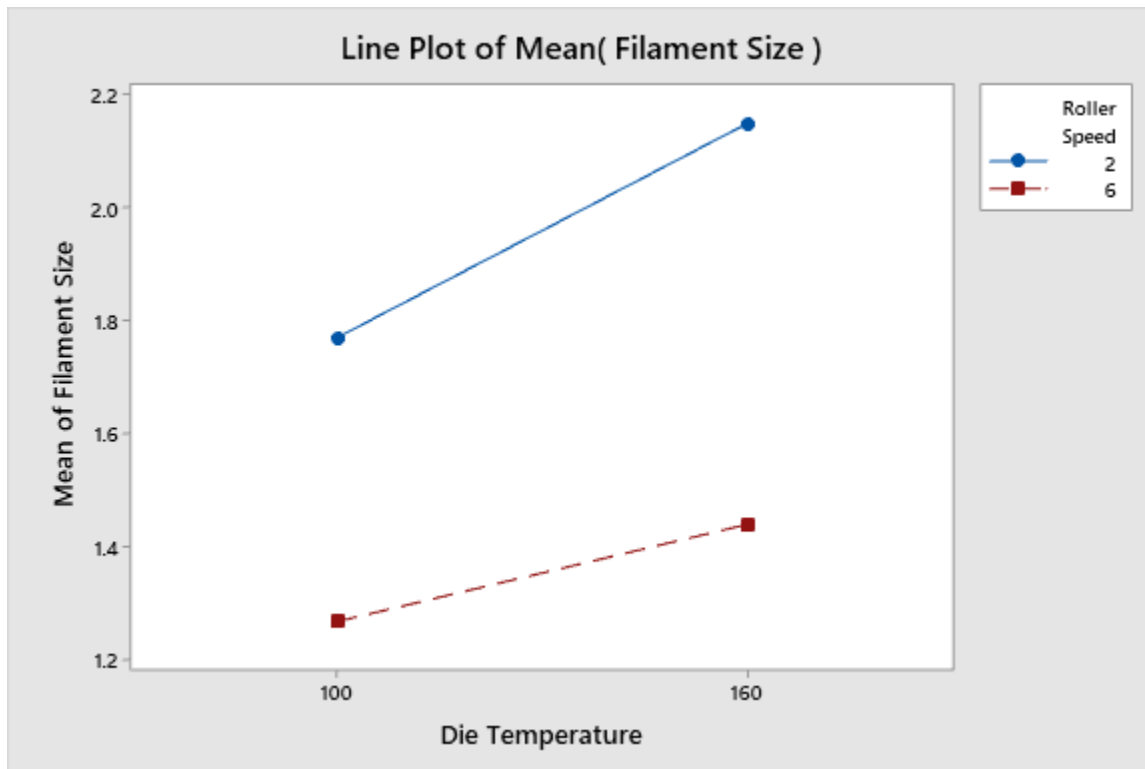


Figure 12 Line plot for mean filament size VS Die Temperature VS Roller Speed

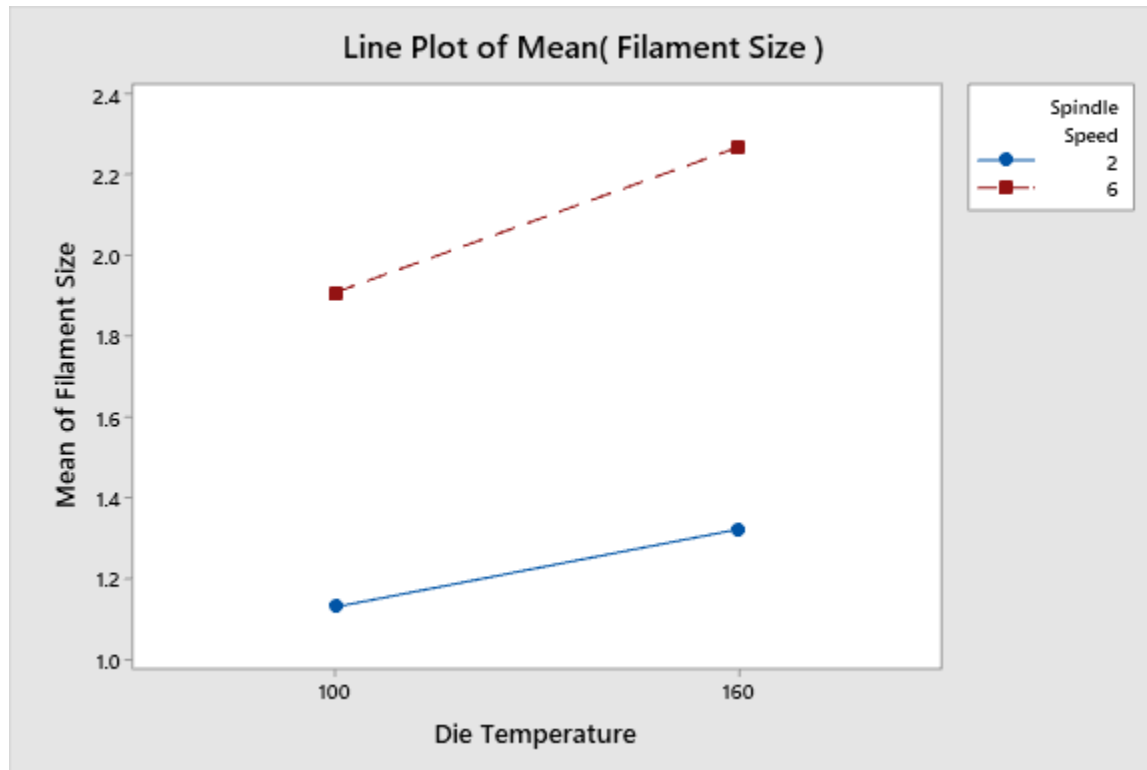


Figure 13 Line plot for mean filament size VS Die Temperature VS Spindle Speed

The regression analysis reveals the relationship between filament size and the four key process parameters:

- Die temperature.
- Roller Speed.
- Spindle Speed.
- Inlet Temperature.

The regression equation:

$$\text{Filament Size} = 1.115 + 0.00460 (\text{Die Temperature}) - 0.1516 (\text{Roller Speed}) + 0.2153 (\text{Spindle Speed}) - 0.00479 (\text{Inlet Temperature})$$

indicates how each factor contributes to changes in filament size. A positive coefficient for die temperature (0.00460) shows a slight increase in filament size with rising temperature, and its statistical significance is supported by a p-value of 0.018.

Roller speed, with a negative coefficient (-0.1516), demonstrates that higher speeds reduce filament size. This factor has a substantial impact on the outcome, confirmed by a highly significant p-value of 0.000.

Spindle speed exhibits the largest influence, with a positive coefficient (0.2153), indicating that increased speeds result in a considerable enlargement of the filament size. The spindle speed's effect is statistically significant, as shown by its p-value of 0.000.

Conversely, inlet temperature shows a minor negative impact (-0.00479), but its high p-value (0.177) suggests that it does not significantly affect filament size.

The **model summary** indicates strong predictive capabilities, with an R-squared value of 91.68%, suggesting that most of the variation in filament size is explained by the model. The adjusted R-squared (88.66%) confirms the robustness of the model, while the predicted R-squared (82.41%) highlights its reliable performance in predicting outcomes. The standard error of regression ($S = 0.199469$) indicates a small average deviation of observed filament sizes from the predicted values, further validating the model's accuracy.

The **ANOVA analysis** reinforces these findings, showing that spindle speed and roller speed are the most significant factors affecting filament size, with F-values of 74.57 and 36.95, respectively, and highly significant p-values of 0.000. Die temperature, with an F-value of 7.67 and a p-value of 0.018, has a moderate but statistically significant impact. In contrast, inlet temperature, with an F-value of 2.08 and a p-value of 0.177, does not significantly contribute to variations in filament size.

Overall, spindle speed and roller speed are identified as critical parameters for achieving precise control over filament dimensions, while die temperature requires careful adjustment to prevent defects. Inlet temperature, being insignificant, can be deprioritized in future optimizations. These insights provide a clear direction for refining the extrusion process to produce consistent and high-quality PLA filaments.

CONCLUSIONS

For also those critical process factors involved in the extrusion process using PLA, full factorial design of experiments (DOE) has offered a great understanding on the effects of the extrusion parameters on the quality of the filaments. This work was able to shed light on the effects of die temperature, roller speed, spindle speed, and inlet temperature on the diameters of the filaments production. The research, with the help of vast experimentation and analysis with the use of Minitab, not only provided an understanding of the most influential factors that affect PLA filament quality but also provided guidelines for stable and high quality of PLA filament production.

Firstly, the regression quantitative analysis also showed the correlation between the selected YSD filament size and the four process parameters. From the regression equation it can be inferred that, the octaves with the highest impact on filament dimensions are spindle speed and roller speed where spindle speed is slightly positive than roller speed. The estimates of the parameters of the models suggest that increases in spindle speeds by a factor of 3 and in filament size lead to major increases in the size of filaments, just as decreases in roller speeds tends to decrease the size of filaments. The presented p-values of 0.000 for both of these parameters prove their high importance in the context of extrusion and RF heating. On the other hand, die temperature had a moderate impact on the size of the filaments and this was affirmed by the p value of 0.018. Inlet temperature, too, did not significantly affect the start-up time as its p-value of 0.177 confirms that it should not be prioritized during time optimization processes.

This was confirmed by the ANOVA analysis Other findings included Therefore, the findings of the study By analyzing the frequency distribution of the scores For this reason, these results supported the major findings Therefore, the results of this study endorsed the above findings and supported them. High F-values for the factors spindle speed (74.57) and roller speed (36.95) indicate why these variables are considered as having a strong influence on the filament size. Still, the die temperature contributes to the results since its F-value equals 7.67. This analysis also pointed out that inlet temperature has no much significance as its F-value was only 2.08 and so it cannot be further discussed. All these results taken together imply that there may be a need to fine tune the spindle speed and roller speed in the extrusion process.

The summary of the model reiterated high reliability of the regression analysis with **91.68% R-squared** established that the model can account for most of the variation in filament size. The **adjusted R-squared of 0.8866** validates the soundness of the model and the predicted **R-squared of 0.8241** testifies the prediction strength of the model. The **standard error of the regression is 0.199469**, it has many zeros proving the practical usability of the model.

In terms of application, therefore, the results of the study suggest specific course of action that can be followed to enhance PLA filament production. The fact that spindle speed and roller speed are ranked highest allows manufacturers to pay attention to these parameters and provide accurate control over them. For instance, the magnitude of spindle speed within the set optimal limit can enhance the size of the filament to the desired size, while the same affects from the roller improves or reduces the size of the filament. The temperature, though the second priority, should also be controlled to avoid such defects as voids in the sample or the irregularity of the filament's surface. As inlet temperature was observed to have little effect on system efficiency, perhaps it does not need to be controlled too closely, thus freeing up more resources.

However, the study has implications with regards to the extrusion technology as an industry, as well. In particular, a full factorial DOE strategy was found to be valuable for identifying broad and narrow effects associated with the studied parameters of the process. This approach minimizes the chances of drawback in understanding the extrusion process and allows determination of favorable conditions with reasonable certainty. These strategic findings outlined in this paper can be applied to other material systems and extrusion processes, particularly in the fabrication of composite filaments and the use of additives towards improving the mechanical characteristics.

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